

Experiment - 1

* AIM :

Design an ER diagram for an Indian e-commerce platform with entities Customer, Product, Order, OrderItem, Seller, Category, Payment, Delivery, Address. Specify primary keys, composite and multivalued attributes, weak entities, specialization of Product, and participation constraints.

* Theory:

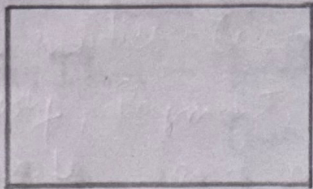
The ER (Entity-Relationship) model, given by Peter Chen, is a conceptual model used to design a database by representing real-world data as entities, attributes, and relationships, independent of any DBMS.

• Key terms —

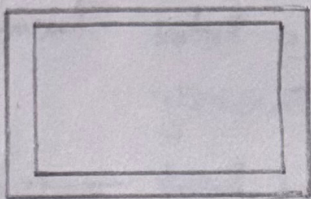
- ▷ Entity — a real-world object with independent existence (Customer, Product).
- ▷ Attribute — a property of an entity (Name of the customer).
- ▷ Relationship — an association between two or more entities (Customer places Order).
- ▷ Cardinality — describing how many instances of one entity relate to another. (1:1, 1:N, or M:N)
- ▷ Participation constraint — total (every instance must participate, shown with a bold/double line) or partial (optional, single line).
- ▷ Weak entity — an entity with no key of its own.

▷ Specialization/generalization — dividing an entity into subclasses with distinct attributes (specialization) or combining entities into a ~~sp~~ superclass (generalisation).

• Notation used:—



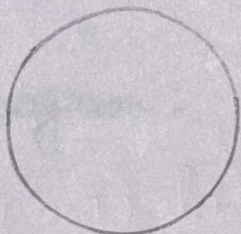
→ Strong entity



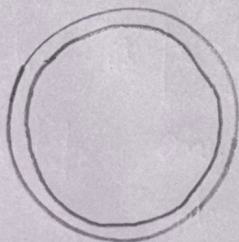
→ Weak entity



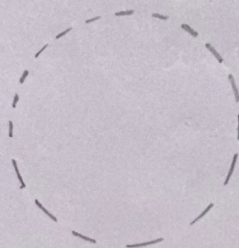
→ Relationship



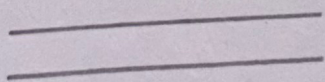
→ Attributes



→ Multi-valued attributes



→ Composite attributes



→ Key attribute

→ Partial key

• Participation constraints :—

- ▷ Order — total participation in places
- ▷ Address — total participation in has
- ▷ OrderItem — total participation in contains and in refers to (weak entity)
- ▷ Product — total participation in sells and classifies
- ▷ Payment — total participation in generates, and Order's side of generates is also total
- ▷ Delivery [total participation in scheduled as
partial participation in delivered to
- ▷ Customer, Seller, Category — partial participation on their of their relationships.

* ER diagram :

